Chapter 2 • Logarithms

Solved Exercise 2.3

1. Find the characteristic of the common logarithm of each number.
(i) $5287=5.287\times10^{3}$, so the characteristic is 3.
(ii) $59.28=5.928\times10^{1}$, so the characteristic is 1.
(iii) $0.0567=5.67\times10^{-2}$, so the characteristic is -2.
(iv) $234.7=2.347\times10^{2}$, so the characteristic is 2.
(v) $0.000049=4.9\times10^{-5}$, so the characteristic is -5.
(vi) $145000=1.45\times10^{5}$, so the characteristic is 5.

2. Find each common logarithm to four decimal places.
(i) $\log43 = \log(4.3\times10)$
$= 1+\log4.3$
$= 1.6335.$
(ii) $\log579 = \log(5.79\times10^{2})$
$= 2+\log5.79$
$= 2.7627.$
(iii) $\log1.982=0.2971.$
(iv) $\log0.0876 = \log(8.76\times10^{-2})$
$= -2+\log8.76$
$= -1.0575.$
(v) $\log0.047 = \log(4.7\times10^{-2})$
$= -2+\log4.7$
$= -1.3279.$
(vi) $\log0.000354 = \log(3.54\times10^{-4})$
$= -4+\log3.54$
$= -3.4510.$

3. Given log 3.177 = 0.5019, find log 3177, log 31.77 and log 0.03177.
$3177 = 3.177 \times 10^{3}$; 31.77
$= 3.177 \times 10^{1}$; 0.03177
$= 3.177 \times 10^{-2}.$
Add 3, 1 and -2 respectively to 0.5019.
(i) 3.5019
(ii) 1.5019
(iii) $-1.4981.$

4. Find x from each given common logarithm.
(i) $\log x=0.0065$
$\Rightarrow x=10^{0.0065}=1.015079...$
1.015
(ii) $\log x=1.192$
$\Rightarrow x=10^{1.192}=15.559656...$
15.56
(iii) $\log x=-3.434$
$\Rightarrow x=10^{-3.434}=0.000368129...$
0.0003681
(iv) $\log x=-1.5726$
$\Rightarrow x=10^{-1.5726}=0.026754695...$
0.02675
(v) $\log x=4.3561$
$\Rightarrow x=10^{4.3561}=22{,}703.8757...$
22,700 to three significant figures
(vi) $\log x=-2.0184$
$\Rightarrow x=10^{-2.0184}=0.009585174...$
0.009585.

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